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Smart Traffic Lights Using Local AI for Adaptive Traffic Management

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***Abstract*—Conventional traffic-light systems generally operate using fixed or predefined timing schedules and may not respond efficiently to rapidly changing traffic conditions. This paper presents a conceptual smart traffic-light system that uses camera-based traffic information and local artificial intelligence processing to support adaptive signal control. Street cameras capture traffic conditions and forward relevant information to a local computing unit positioned near the intersection. The local unit estimates traffic conditions and generates a control decision for the traffic-light controller. Selected information may also be forwarded to a cloud server for storage and wider analysis. The proposed architecture is additionally designed to support emergency vehicle detection, with the objective of coordinating nearby signals to facilitate a clear route. The system architecture, mathematical model, flowchart, algorithm, and evaluation framework are presented as a prototype research design.**

***Index Terms*—Smart traffic lights, local artificial intelligence, computer vision, adaptive traffic control, traffic management, emergency vehicle detection.**

I. Introduction

Traffic congestion is an important problem in urban transportation systems. Conventional traffic signals commonly use predefined timing schedules. Although such systems are simple to implement, fixed timing may not respond effectively to sudden changes in traffic demand.

Artificial intelligence and computer vision have been investigated as approaches for adaptive traffic signal control. Rasheed *et al.* reviewed the application of deep reinforcement learning to traffic signal control and discussed its potential for addressing complex traffic conditions [1].

Camera-based traffic analysis has also been investigated for adaptive signal control. Gorodokin *et al.* studied machine-vision-based traffic monitoring and adaptive traffic light control using vehicle and queue information [2].

However, a system that depends entirely on a remote server may introduce communication requirements and additional delay. Therefore, this paper considers a local-AI architecture in which traffic information is processed close to the intersection.

The main objectives of the proposed system are:

- To monitor traffic conditions using cameras.
- To process traffic information locally.
- To support adaptive traffic-light decisions.
- To reduce unnecessary dependence on continuous cloud processing.
- To provide a framework for emergency vehicle detection and route coordination.

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II. Related Work

Research in intelligent traffic signal control has explored several approaches, including reinforcement learning and computer vision.

Rasheed *et al.* presented a review of deep reinforcement learning techniques for traffic signal control. The study discusses traffic signal control attributes, learning architectures, simulation platforms, and challenges associated with applying deep reinforcement learning to traffic systems [1].

Gorodokin *et al.* investigated adaptive traffic-light control using machine vision. Their approach considered traffic-flow parameters such as vehicle information and queue conditions when determining signal timing [2].

These studies demonstrate the relevance of intelligent traffic monitoring and adaptive signal control. The proposed work focuses specifically on a local-processing architecture that can be used as a prototype for adaptive traffic management.

III. Proposed System

The proposed system consists of four major stages:

1) Street cameras capture traffic conditions.
2) Local computing units process the camera information.
3) A traffic-control decision is generated from the processed information.
4) The decision is transmitted to the traffic-light controller.

Selected traffic information may additionally be transmitted to a cloud server for storage and large-scale analysis.

The proposed architecture therefore follows the general sequence:

$$\text{Camera} \rightarrow \text{Local AI} \rightarrow \text{Decision} \rightarrow \text{Traffic Signal}. \quad (1)$$

IV. System Architecture

The system architecture contains street cameras, local computing units, a traffic-light controller, and an optional cloud server.

The cameras provide visual information about vehicles and traffic conditions. The local computing unit processes the incoming information and estimates parameters such as traffic density or queue conditions.

The resulting decision is transmitted to the traffic-light controller. The cloud server is not required for every real-time decision and can instead be used for storage, historical analysis, or centralized monitoring.

This local-processing concept is motivated by the need to reduce unnecessary communication of raw camera data and to support responsive decision-making.

The conceptual architecture is illustrated in Fig. 1.

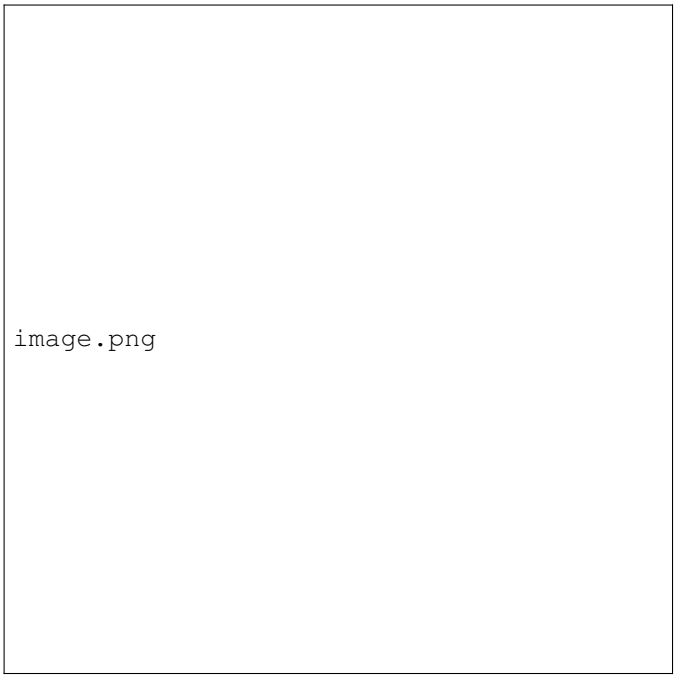


Fig. 1. Conceptual architecture of the proposed local-AI traffic management system.

V. System Components

The major functional components of the proposed system are listed in Table I.

TABLE I
MAJOR COMPONENTS OF THE PROPOSED SYSTEM

Component	Function
Street Camera	Captures traffic conditions and vehicle information.
Local Computing Unit	Processes camera data and estimates traffic conditions.
Traffic Controller	Receives control decisions and operates traffic signals.
Cloud Server	Stores selected information for historical analysis and monitoring.
Emergency Vehicle Module	Detects or receives information about an emergency vehicle and supports route coordination.

VI. Traffic Density Model

A simple traffic-density measure can be defined using the number of detected vehicles and the observation region.

Let N represent the number of detected vehicles and A represent the observation area. The estimated vehicle density can be expressed as

$$D = \frac{N}{A} \tag{2}$$

where D represents vehicle density.

The local controller can use the estimated density together with other traffic parameters to determine an appropriate signal phase.

For example, a simple normalized priority measure can be represented as

$$P_i = w_1 D_i + w_2 Q_i + w_3 E_i \tag{3}$$

where:

- D_i is the traffic density for lane or approach i,
- Q_i is the queue-related parameter,
- E_i represents emergency-vehicle priority,
- w_1, w_2, and w_3 are weighting parameters.

The weights are design parameters and must be determined experimentally for a real implementation.

VII. Traffic Data Representation

The traffic conditions detected from multiple approaches can be represented using an array:

$$\mathbf{D} = \begin{bmatrix} D_1 & D_2 & D_3 & D_4 \end{bmatrix} \tag{4}$$

where D_1 through D_4 represent the estimated traffic densities of four approaches.

The corresponding queue values can be represented as

$$\mathbf{Q} = \begin{bmatrix} Q_1 & Q_2 & Q_3 & Q_4 \end{bmatrix} \tag{5}$$

A state matrix can also be used to represent the condition of the intersection:

$$\mathbf{S} = \begin{bmatrix} D_1 & Q_1 & E_1 \\ D_2 & Q_2 & E_2 \\ D_3 & Q_3 & E_3 \\ D_4 & Q_4 & E_4 \end{bmatrix} \tag{6}$$

VIII. System Flowchart

The operational flow of the proposed system is shown in Fig. 2.

IX. Proposed Algorithm

The basic operating procedure is given in Algorithm 1.

Algorithm 1 Local-AI Traffic Signal Control

```
Initialize camera and traffic controller
Capture traffic information
Process camera information locally
Estimate vehicle density and queue conditions
if emergency vehicle is detected then
    Generate emergency-route priority
else
    Calculate traffic priority
    Determine appropriate signal phase
end if
Send decision to traffic controller
Update traffic signal
Store selected information if required
Repeat the process
```

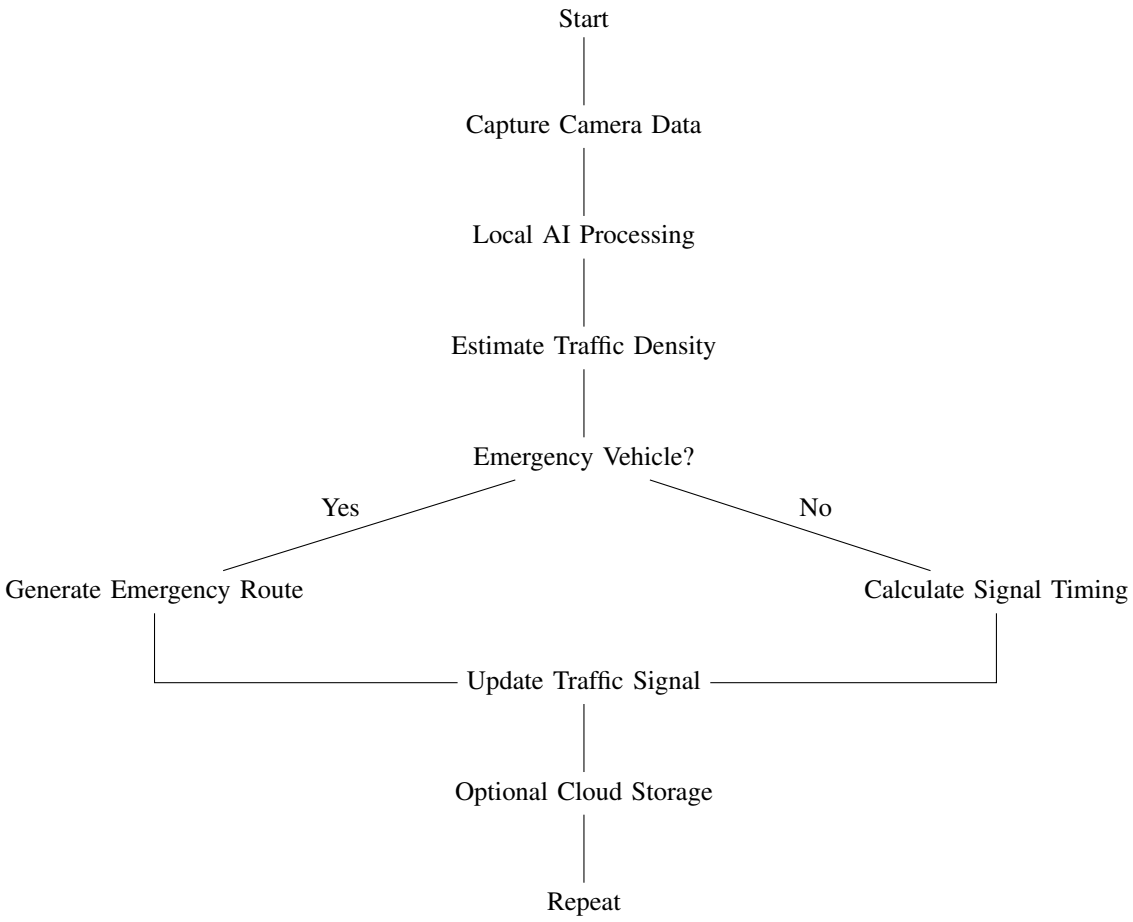


Fig. 2. Flowchart of the proposed local-AI traffic management system.

prototype.jpg

Fig. 3. Prototype implementation of the proposed system.

X. Prototype Visualization

An actual prototype image can be inserted using the following figure environment.

If the image file is named `prototype.jpg`, it can be included as follows:

XI. Evaluation Framework

The proposed system should be evaluated using measurable parameters rather than unsupported performance claims.

The following parameters can be considered:

- Detection accuracy.
- Average signal-decision time.
- Vehicle waiting time.
- Queue length.
- Emergency-route clearance time.
- Communication data volume.

For example, average waiting time can be calculated as

$$T_{\text{avg}} = \frac{1}{n} \sum_{i=1}^{n} T_i \tag{7}$$

where T_i is the waiting time of vehicle i and n is the number of observed vehicles.

XII. Results and Discussion

This sample paper does not claim measured experimental results because no experimental dataset or prototype measurements were provided.

For an actual implementation, the proposed system should be tested under different traffic conditions, including low, medium, and high traffic density.

Emergency-vehicle scenarios should also be evaluated separately. Measurements such as detection accuracy, decision time, queue length, and emergency-route clearance time can then be reported.

A results table can be created as follows:

TABLE II
Example Format for Experimental Results

Scenario	Vehicles	Avg. Wait	Decision Time
Low Traffic	–	–	–
Medium Traffic	–	–	–
High Traffic	–	–	–
Emergency	–	–	–

The dashes in Table II must be replaced by actual experimental measurements. They are intentionally not filled with fabricated values.

XIII. Discussion

Local processing can reduce the need to continuously transmit raw camera information to a remote server. It can also allow the traffic controller to receive decisions from a computing unit located near the intersection.

However, the effectiveness of the proposed system depends on several factors, including camera quality, traffic detection accuracy, processing capability, communication reliability, and the control algorithm.

The system should therefore be evaluated using real traffic data or a validated simulation environment before conclusions about performance are made.

XIV. Limitations

The proposed system is presented as a prototype research architecture and has several limitations.

First, camera-based detection can be affected by lighting, weather, occlusion, and camera placement.

Second, the emergency-vehicle module requires reliable detection or identification of emergency vehicles.

Third, the proposed mathematical model uses simplified traffic parameters. A real deployment would require a more detailed traffic model and extensive testing.

Finally, no numerical performance improvement is claimed in this paper because experimental measurements have not been provided.

XV. Conclusion

This paper presented a conceptual smart traffic-light system based on local artificial intelligence and camera-based traffic monitoring.

The proposed architecture processes traffic information near the intersection and uses the resulting information to support adaptive signal decisions. An optional cloud layer can be used for storage and wider analysis rather than requiring all raw camera information to be processed remotely.

The architecture also provides a framework for emergency vehicle detection and route coordination. Future work should include prototype implementation, real-world or simulated traffic datasets, emergency-vehicle detection, and quantitative evaluation using waiting time, queue length, detection accuracy, and decision time.

References

[1] F. Rasheed, K. L. A. Yau, R. M. Noor, C. Wu, and Y.-C. Low, "Deep reinforcement learning for traffic signal control: A review," *IEEE Access*, vol. 8, pp. 208 016–208 044, 2020.

[2] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, "Optimization of adaptive traffic light control modes based on machine vision," *Transportation Research Procedia*, vol. 57, pp. 241–249, 2021.